Unit 11: Mechanical Vibrations and Oscillators

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through simple harmonic motion setups, free undamped initial-value problems, the amplitude-phase collapse, Hooke's law calibration, the three damping regimes sorted by the discriminant, the critical damping coefficient, resonance identification, and a coupled-oscillator capstone. Solutions follow in Part III.

1. **(SHM setup)** For $2x'' + 18x = 0$, find the natural angular frequency ω_0, the period T, and the ordinary frequency f.

2. **(Free undamped IVP)** Solve $x'' + 9x = 0$ with $x(0) = 0$, $x'(0) = 6$.

3. **(Amplitude-phase form)** Write $x(t) = 5\cos 2t + 12\sin 2t$ as a single sinusoid $C\cos(2t - \phi)$; give the amplitude C and phase ϕ.

4. **(Hooke's law calibration)** A weight of $mg = 10$ N stretches a spring by $L = 0.2$ m. Find the stiffness k, and with $g = 10$ m/s^2 find the resulting natural frequency ω_0.

5. **(Damping — overdamped)** Solve $x'' + 5x' + 6x = 0$, evaluate the discriminant, and name the damping regime.

6. **(Damping — critically damped)** Solve $x'' + 2x' + x = 0$ and identify the regime from the discriminant.

7. **(Damping — underdamped)** Solve $x'' + 2x' + 5x = 0$, name the regime, and state the quasi-frequency.

8. **(Critical damping coefficient)** For $m = 1$ and $k = 9$, find the damping coefficient c that makes the system critically damped.

9. **(Resonance identification)** For $x'' + 25x = \cos(\omega t)$, find the forcing frequency ω that produces resonance, and compute the resonant particular solution at that frequency.

10. **(Coupled oscillators, capstone)** For $x_1'' = -2x_1 + x_2$ and $x_2'' = x_1 - 2x_2$, find the normal-mode frequencies from the eigenvalues $\lambda = -1, -3$, identify the mode shapes, and write the general solution.

Part III — Complete Solutions

Solution 1. Divide by the mass 2 to reach standard form:

$$2x'' + 18x = 0 \implies x'' + 9x = 0 \implies \omega_0^2 = 9,\ \omega_0 = 3.$$

Then the period and ordinary frequency follow from ω_0:

$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{3}, \qquad f = \frac{\omega_0}{2\pi} = \frac{3}{2\pi}.$$

Reading ω_0 off the *undivided* coefficient 18 is the classic error; the mass divides out first.

Solution 2. The roots of $r^2 + 9 = 0$ are $\pm 3i$, so $x = c_1 \cos 3t + c_2 \sin 3t$. Apply the data:

$$x(0) = c_1 = 0, \qquad x'(t) = 3c_2 \cos 3t, \quad x'(0) = 3c_2 = 6 \ \Rightarrow\ c_2 = 2.$$

$$x(t) = 2\sin 3t.$$

The derivative pulls down a factor of $\omega_0 = 3$, so c_2 is $6/3 = 2$, not 6.

Solution 3. Combine the coefficients into one amplitude:

$$C = \sqrt{c_1^2 + c_2^2} = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

The phase satisfies $\tan\phi = c_2/c_1 = 12/5$ with both components positive (first quadrant):

$$\phi = \arctan \tfrac{12}{5} \approx 1.176 \text{ rad}, \qquad x(t) = 13\cos(2t - 1.176).$$

Solution 4. The static balance $mg = kL$ gives the stiffness directly:

$$k = \frac{mg}{L} = \frac{10}{0.2} = 50 \text{ N/m}.$$

With $g = 10$ m/s^2 the weight $mg = 10$ N gives mass $m = 1$ kg, so

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{50}{1}} = 5\sqrt{2} \approx 7.07 \text{ rad/s}.$$

Solution 5. The characteristic equation factors:

$$r^2 + 5r + 6 = (r+2)(r+3) = 0 \ \Rightarrow\ r = -2,\ -3.$$

The discriminant $\Delta = c^2 - 4mk = 25 - 24 = 1 > 0$ confirms two distinct real roots, so the motion is **overdamped** — pure decay, no oscillation:

$$x(t) = c_1 e^{-2t} + c_2 e^{-3t}.$$

Solution 6. The characteristic equation is a perfect square:

$$r^2 + 2r + 1 = (r+1)^2 = 0 \ \Rightarrow\ r = -1 \text{ (double)}.$$

Here $\Delta = 4 - 4 = 0$, the boundary case, so the system is **critically damped** and the repeated root carries a factor of t:

$$x(t) = (c_1 + c_2 t)\, e^{-t}.$$

Solution 7. The quadratic formula on $r^2 + 2r + 5 = 0$ gives complex roots:

$$r = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i.$$

With $\Delta = 4-20 = -16 < 0$ the motion is **underdamped**: a decaying envelope times an oscillation,

$$x(t) = e^{-t}\big(c_1 \cos 2t + c_2 \sin 2t\big).$$

The quasi-frequency is $\mu = 2$, just below the undamped value $\omega_0 = \sqrt{5} \approx 2.236$.

Solution 8. Critical damping is the boundary $\Delta = 0$, i.e. $c^2 = 4mk$, so

$$c = 2\sqrt{mk} = 2\sqrt{(1)(9)} = 2 \cdot 3 = 6.$$

Any $c > 6$ overdamps the system and any $c < 6$ underdamps it.

Solution 9. From $x'' + 25x = 0$ the natural frequency is $\omega_0 = \sqrt{25} = 5$, so resonance occurs at

$$\omega = \omega_0 = 5.$$

At resonance the bare trial $A\cos 5t + B\sin 5t$ already solves the homogeneous equation, so it gains a factor of t: try $x_p = t(A\cos 5t + B\sin 5t)$. Differentiating twice,

$$x_p'' = -25\,t(A\cos 5t + B\sin 5t) + 10(-A\sin 5t + B\cos 5t),$$

so $x_p'' + 25x_p = 10(-A\sin 5t + B\cos 5t)$. Matching to $\cos 5t$:

$$10B = 1, \quad -10A = 0 \;\Rightarrow\; A = 0,\; B = \tfrac{1}{10}, \qquad x_p(t) = \frac{t}{10}\sin 5t.$$

The t out front is the signature of resonance: the amplitude grows without bound.

Solution 10 (Capstone). In matrix form $\mathbf{x}'' = A\mathbf{x}$ with

$$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}, \qquad \lambda = -1,\, -3.$$

Each mode frequency comes from $\omega^2 = -\lambda$:

$$\omega_1 = \sqrt{-(-1)} = 1, \qquad \omega_2 = \sqrt{-(-3)} = \sqrt{3}.$$

The eigenvectors give the mode shapes. For $\lambda = -1$, $(A+I)\mathbf{v} = 0$ yields $\mathbf{v}_1 = (1,1)^\mathsf{T}$, the **in-phase** mode (lower frequency); for $\lambda = -3$, $(A+3I)\mathbf{v} = 0$ yields $\mathbf{v}_2 = (1,-1)^\mathsf{T}$, the **out-of-phase** mode (higher frequency, since it stretches the coupling spring more). The general motion is a superposition of the two modes:

$$\mathbf{x}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}\big(a_1 \cos t + b_1 \sin t\big) + \begin{pmatrix} 1 \\ -1 \end{pmatrix}\big(a_2 \cos \sqrt{3}\,t + b_2 \sin \sqrt{3}\,t\big).$$

Unit 11 Key Takeaway

Read the motion straight off the coefficients. The natural frequency $\omega_0 = \sqrt{k/m}$ and the collapse to $C\cos(\omega_0 t - \phi)$ govern free motion; the discriminant $c^2 - 4mk$ sorts damping into overdamped (> 0), critically damped ($= 0$, $c = 2\sqrt{mk}$), and underdamped (< 0); resonance at $\omega = \omega_0$ injects the t factor that drives $x_p = \frac{t}{10}\sin 5t$ to grow without bound; and the eigenvalues of $\mathbf{x}'' = A\mathbf{x}$ deliver the normal-mode frequencies $\omega = \sqrt{-\lambda}$ whose superposition is the full coupled motion.